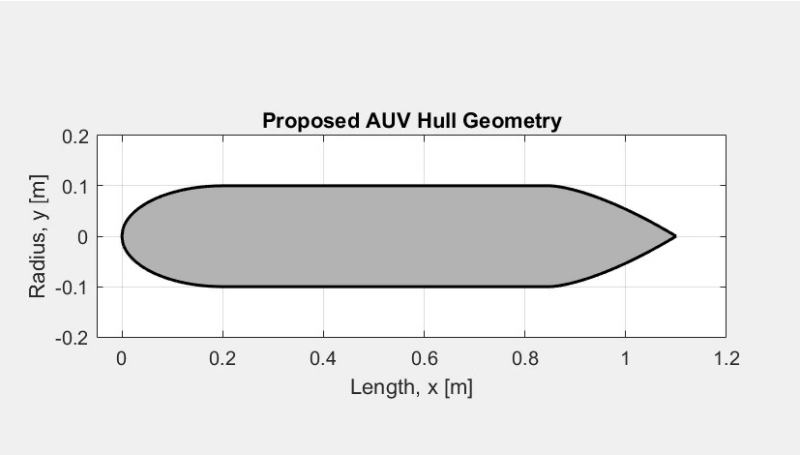
Autonomous Underwater Vehicle

Fall 2026 · In progress · Individual Phase 1, followed by team development

For Fluid Dynamics, I developed an underwater-vehicle hull concept for navigating narrow passages. My individual Phase 1 work covered vehicle precedents, preliminary geometry, hydrostatics, buoyancy and a qualitative prediction of flow around the hull.



Preliminary hull profile from my individual Phase 1 study.

My contribution

My individual Phase 1 work compared streamlined vehicles, multi-thruster systems and underwater gliders before proposing a symmetric 1.10 m-long hull with a 0.20 m maximum diameter. Rear propulsion, auxiliary thrusters and forward sensing are proposed for navigating confined passages.

Preliminary hydrostatics

For an idealized fully submerged hull in freshwater at 1,000 kg/m³ and a 5 m depth, the archived calculations report 49.1 kPa external gauge pressure, 27.2 L displacement, 267 N buoyant force and 27.2 kg neutral-buoyancy mass. The stated longitudinal center of buoyancy is 0.501 m aft of the nose. The original hull integration files are needed to reproduce its volume and center.

Status and limits

Individual literature review, hull concept and preliminary calculations are complete; the team phase is in progress in Fall 2026. The flow sketch is a qualitative prediction, not CFD or a measurement. No constructed vehicle, autonomous navigation or underwater-test result is claimed.

Evidence basis: Student-supplied engineering portfolio, 27 September 2026; associated résumé.